

WASP-43b

R-1084

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_160314 · created 2026-07-30T16:00:36+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457461.6849 +/- 0.00072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.012
Transit depth (Rp/Rs)^2	1.63 +/- 0.31 [%]
Orbital Inclination (inc)	84.61 +/- 1.2
Airmass coefficient 1 (a1)	0.943 +/- 0.013
Airmass coefficient 2 (a2)	0.0379 +/- 0.0041
Scatter in the residuals of the lightcurve fit is	1.33 %
Best Comparison Star	#3 - [171, 342]
Optimal Aperture	3.49
Optimal Annulus	15.58
Transit Duration (day)	0.055 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.128 $\pm$ 0.012 vs 0.1594 (deviation 2.12 σ)
Duration fit vs published	0.055 d vs 0.0512 d (deviation 1.21 σ)
Mid-time O–C	2.6 min
Depth SNR	8.6
Residual scatter	1.33 %

Light curve

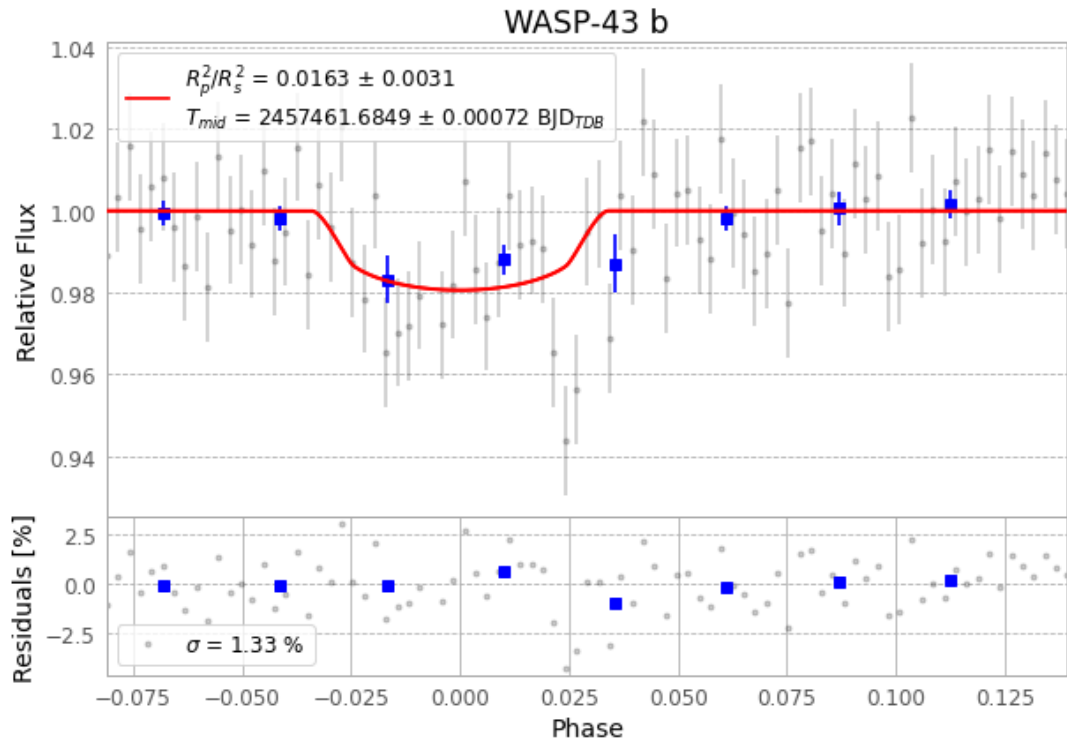


Figure 1 — FinalLightCurve_WASP-43 b_2016-03-13.png (SHA-256 418fc4798ef81392...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [150, 216], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7ce64109aeae21c3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

87 FITS frames (57 MB), WASP-43160314024213.FITS → WASP-43160314070012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-160314.log (b26f0a9acca1...), FinalLightCurve_WASP-43 b_2016-03-13.png (418fc4798ef8...), FinalLightCurve_WASP-43 b_2016-03-13.csv (92c298482f76...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 16:00Z.